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Online Test 1, 2021 (Test)

 INSTRUCT

MJ (Matshisa) Lego

University of Preto

Current Score

QUESTION

1

2

3

4

5

6

7

8

9

10

Due Date

SAT, MAY 22, 2021**2:00 PM GMT+2**[Request Extension](#)

Instructions

Answer all questions. You have two submission opportunities. The duration of the test is 90 minutes; and runs from 14:00 on 20 May (Thurs) to 14:00 on 22 May (Sat) 2021. Submit a PDF file of the filled-in Online Test 1 and a scan of your workings in the "Semester and Online Tests Folder" link, under the "Online Test 1 Submission" item on ClickUP.

Description

Assignment Submission & Scoring

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

1.

[DETAILS](#)[PREVIOUS ANSWERS](#)

SERPSE9 1.P.027.MI.FB.

[MY NOTES](#)[ASK YOUR TEACHER](#)

One gallon of paint (volume = $3.78 \times 10^{-3} \text{ m}^3$) covers an area of 34.0 m^2 . What is the thickness of the fresh paint on the wall?

 m

2.

DETAILS

PREVIOUS ANSWERS

MY NOTES

ASK YOUR TEACHER

A car traveling 56 km/h is 24.0 m from a barrier when the driver slams on the brakes. The car hits the barrier 2.20 s later.

(a) Assuming constant acceleration, what was the magnitude of the car's acceleration before impact?

m/s²

(b) How fast was the car traveling at impact?



m/s

3.

DETAILS

PREVIOUS ANSWERS

MY NOTES

ASK YOUR TEACHER



At $t=0$, you stand at the origin and throw a football at an angle Θ_0 with respect to the $+x$ -axis. The ball reaches a maximum height in its trajectory and then at a time of **5.64 s** crashes into a wall that is **324.2 m** from the origin. The ball leaves your hand with a speed of **70.6 m/s**. MAKE AN ACCURATE DIAGRAM OF THE MOTION AND INITIAL INFORMATION BEFORE GOING ON

(a) Find the horizontal component of the football's initial velocity.

✓ m/s

(b) Find the vertical component of the football's initial velocity.

m/s

(c) Find the angle of the throw, Θ_0 . ✗ degrees

(d) When does the ball reach maximum height? ✗ s

(e) What is the maximum height of the football? ✗ m

(f) At what height does the football hit the wall? ✗ m

4.

DETAILS

PREVIOUS ANSWERS

MY NOTES

ASK YOUR TEACHER

A jogger runs eastward in a straight line with an average speed of 2.11 m/s for 2.11 min and then continues with an average speed of 2.13 m/s for 2.27 min . (Hint: time is in minutes and velocity is in meters/second)

(a) What is her total displacement (in m)?



m

(b) What is her average velocity (in m/s) during this entire jog?



m/s

5.

DETAILS

PREVIOUS ANSWERS

SERPSE10 2.A.OP.037.

MY NOTES

ASK YOUR TEACHER

A cab driver is driving a car along a straight road at a constant speed of 21.5 m/s . Just as he passes a law officer parked along the side of the road, the officer starts to accelerate 0.50 m/s^2 to overtake him. Assume that the officer maintains this acceleration.

- (a) From the moment the cab driver passes the officer, determine the time it takes the officer to reach the cab driver (in s).

 s

- (b) Find the speed of the officer as he overtakes the cab driver (in m/s).

 m/s

- (c) Find the total displacement of the officer as he overtakes the cab driver (in m).

 m

- (d) **What If?** Determine the answers to parts (a) through (c) of the problem, instead of accelerating at the same instant the car passes him, the officer instead begins his acceleration 1.00 s after the car passes him. (Enter the time in s, speed in m/s, and displacement in m.)

time sspeed m/sdisplacement m

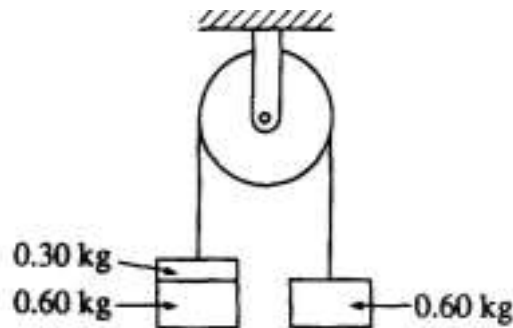
6.

DETAILS

PREVIOUS ANSWERS

MY NOTES

ASK YOUR TEACHER



Two 0.60 kilogram objects are connected by a thread that passes over a light, frictionless pulley, as shown above. The objects are initially held at rest. If a third object with a mass of 0.30 kilogram is added on top of one of the 0.60-kilogram objects as shown and the objects are released, the magnitude of the acceleration of the 0.30-kilogram object is most nearly

- ☐ 3.0 m/s²
- ☐ 6.0 m/s²
- ☐ 10.0 m/s²
- ☐ 1.0 m/s²
- ☒ 2.0 m/s²



7.

DETAILS

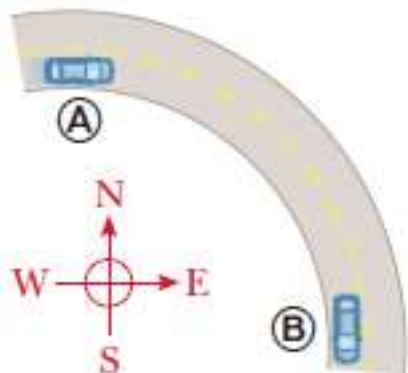
PREVIOUS ANSWERS

SERPSE10 6.A.P.024.

MY NOTES

ASK YOUR TEACHER

A car travels clockwise at constant speed around a circular section of a horizontal road as shown in the aerial view of the figure below.



i

(a) Find the directions of its velocity and acceleration at position **A**.

direction of velocity

east



direction of acceleration

south



(b) Find the directions of its velocity and acceleration at position **B**.

direction of velocity

south



direction of acceleration

west



8.

DETAILS

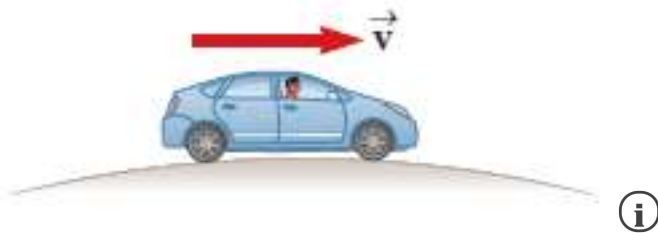
PREVIOUS ANSWERS

SERPSE10 6.A.P.027.

MY NOTES

ASK YOUR TEACHER

A car of mass m passes over a hump in a road that follows the arc of a circle of radius R as shown in the figure below.



(a) If the car travels at a speed v , what force does the road exert on it as the car passes the highest point of the hump? (Use any variable or symbol stated above along with the following as necessary: g .)

 $n =$

 $m \cdot g + m \cdot v^2 / R$

✗

(b) What is the maximum speed the car can have without losing contact with the road as it passes this highest point? (Use any variable or symbol stated above along with the following as necessary: g .)

 $v_{\max} =$

 $\sqrt{R \cdot g}$



9.

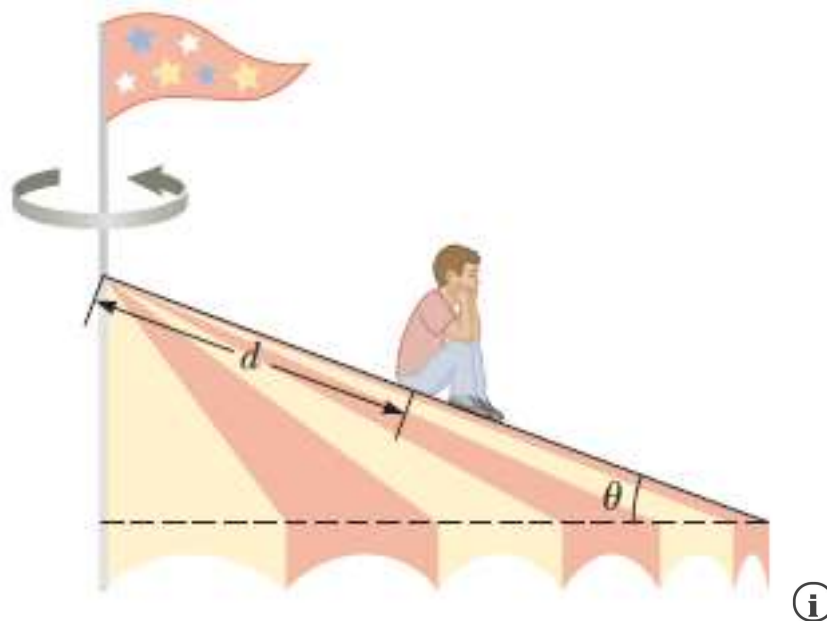
DETAILS

SERPSE10 6.A.P.032.

MY NOTES

ASK YOUR TEACHER

Why is the following situation impossible? A mischievous child goes to an amusement park with his family. On one ride, after a severe scolding from his mother, he slips out of his seat and climbs to the top of the ride's structure, which is shaped like a cone with its axis vertical and its sloped sides making an angle $\theta = 20.0^\circ$ with the horizontal as shown in the figure below. This part of the structure rotates about the vertical central axis when the ride operates. The child sits on the sloped surface at a point $d = 5.32$ m down the sloped side from the center of the cone and pouts. The coefficient of static friction between the boy and the cone is $\mu_s = 0.700$. The ride operator does not notice that the child has slipped away from his seat and so continues to operate the ride. As a result, the sitting, pouting boy rotates in a circular path at a speed of 3.75 m/s.



The centrifugal force is bigger than the static friction force and thus would the kid slip down the slope

10.

DETAILS

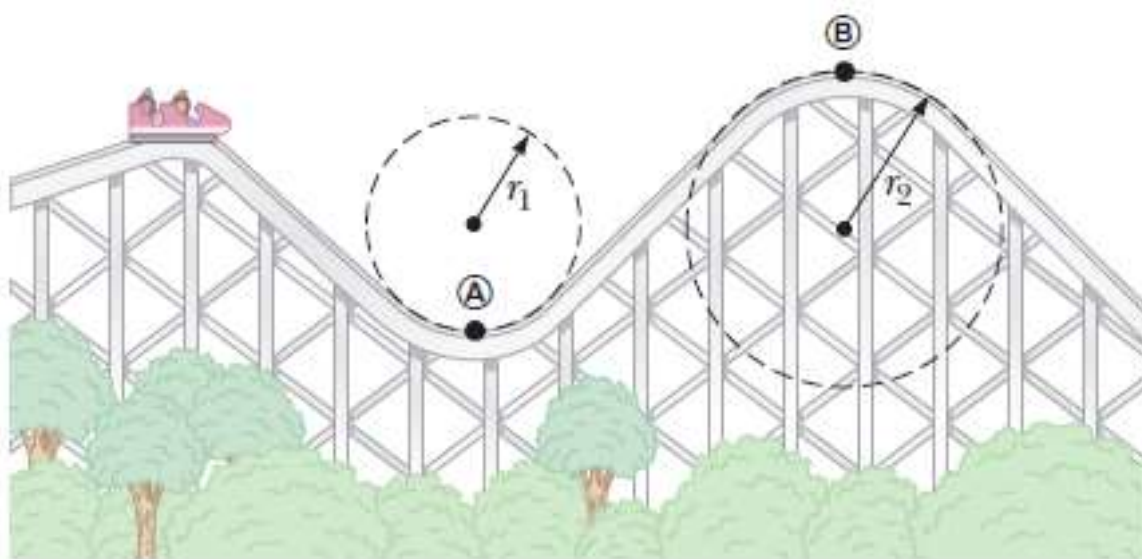
PREVIOUS ANSWERS

SERPSE10 6.A.OP.021.

MY NOTES

ASK YOUR TEACHER

- (a) A roller-coaster car has a total mass (including passengers) of **525** kg. Sit the car is a passenger with a mass of **52.0** kg. The car reaches the lowest of a circular arc in the track, point A in the figure below, moving at a speed **17.0** m/s. The radius of the arc is $r_1 = \mathbf{25.0}$ m.



What is the magnitude (in N) and direction of the force that the seat exerts on the passenger at point A?

magnitude

11214



N

direction

upwards



- (b) **What If?** If the car has the same speed at point A as in part (a), what would the radius (in m) of the track have to be for the force of the seat on the passenger at this point to be **twice** the passenger's weight?

32.7



m

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